LAB 9 - Balanced Wheatstone Bridge
TECH 60 - Sec 14
F26
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Objective:
The objective of this lab is to understand the Wheatstone Bridge and the relationship between voltages and currents in a balanced Wheatstone Bridge circuit. We find the potentiometer resistance that balances the bridge and compare hand calculations, a Multisim simulation, and a breadboard circuit, both unloaded and with a 10 kΩ load between points A and B.

Equipment/Materials:
Solderless Breadboard
Keysight EDU36311A, Triple Output, Programmable DC Power Supply
Keysight EDU34450A 5½ Digit Digital Multimeter
Multisim software running on PC
1500 Ω resistors (Quantity 1)
750 Ω resistors (Quantity 1)
470 Ω resistors (Quantity 1)
390 Ω resistors (Quantity 1)
10 kΩ resistors (Quantity 1)
1 kΩ potentiometer
Jumper wires

Part I. Hand Calculations
$V_s = 9$ V
$R_1 = R_{POT}$ (1 kΩ potentiometer)
$R_2 = 750\ \Omega$
$R_3 = 470\ \Omega$
$R_4 = 390\ \Omega$

Balance condition (the bridge is balanced when $V_A = V_B$)
$V_A = V_s \times R_3/(R_1 + R_3)$
$V_B = V_s \times R_4/(R_2 + R_4)$
$V_A = V_B$ gives $R_3/(R_1 + R_3) = R_4/(R_2 + R_4)$, which simplifies to $R_1/R_3 = R_2/R_4$

Value of R_{POT} at balance
$R_{POT} = R_1 = R_2 \times R_3/R_4$
$R_{POT} = 750\ \Omega \times 470\ \Omega/390\ \Omega = 352{,}500/390 = 903.85\ \Omega$

Voltage at A (voltage divider with R_1 and R_3)
$V_A = V_s \times R_3/(R_1 + R_3)$
$V_A = 9\text{ V} \times 470\ \Omega/(903.85\ \Omega + 470\ \Omega) = 4230/1373.85 = 3.079$ V

Voltage at B (voltage divider with R_2 and R_4)
$V_B = V_s \times R_4/(R_2 + R_4)$

V_B = 9 V × 390 Ω/(750 Ω + 390 Ω) = 3510/1140 = 3.079 V

Output voltage
V_OUT = V_A - V_B
V_OUT = 3.079 V - 3.079 V = 0 V

Check (resistor ratios)
R_1/R_3 = 903.85/470 = 1.923
R_2/R_4 = 750/390 = 1.923
The ratios are equal, so the bridge is balanced.

Part II. Circuit simulation using Multisim

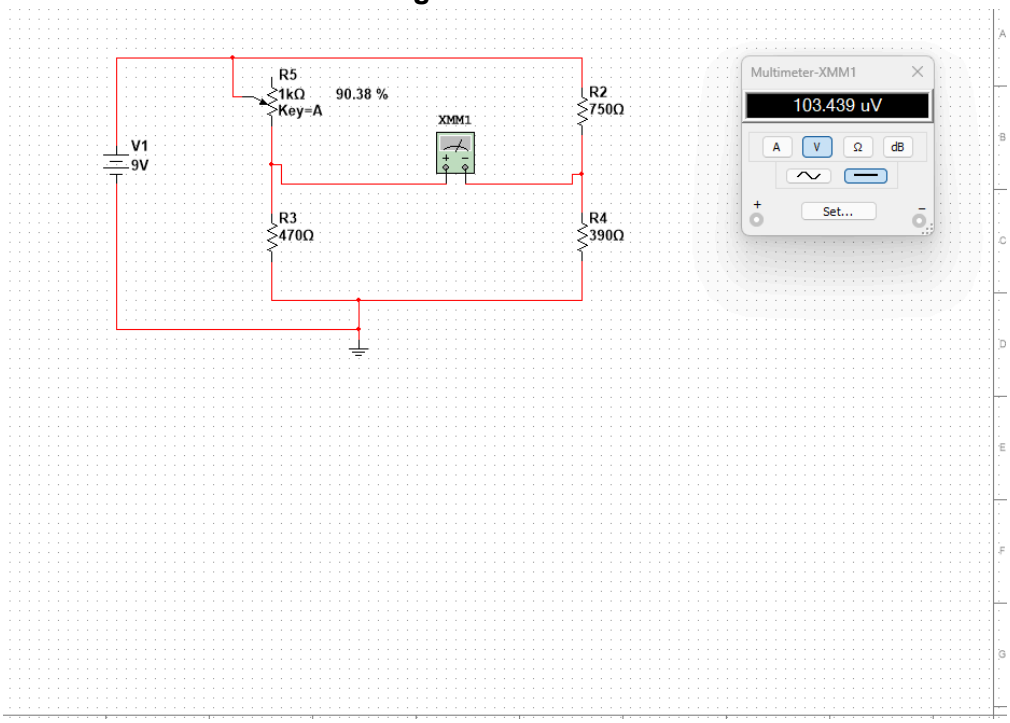


Figure 1. Balanced Wheatstone Bridge circuit in Multisim

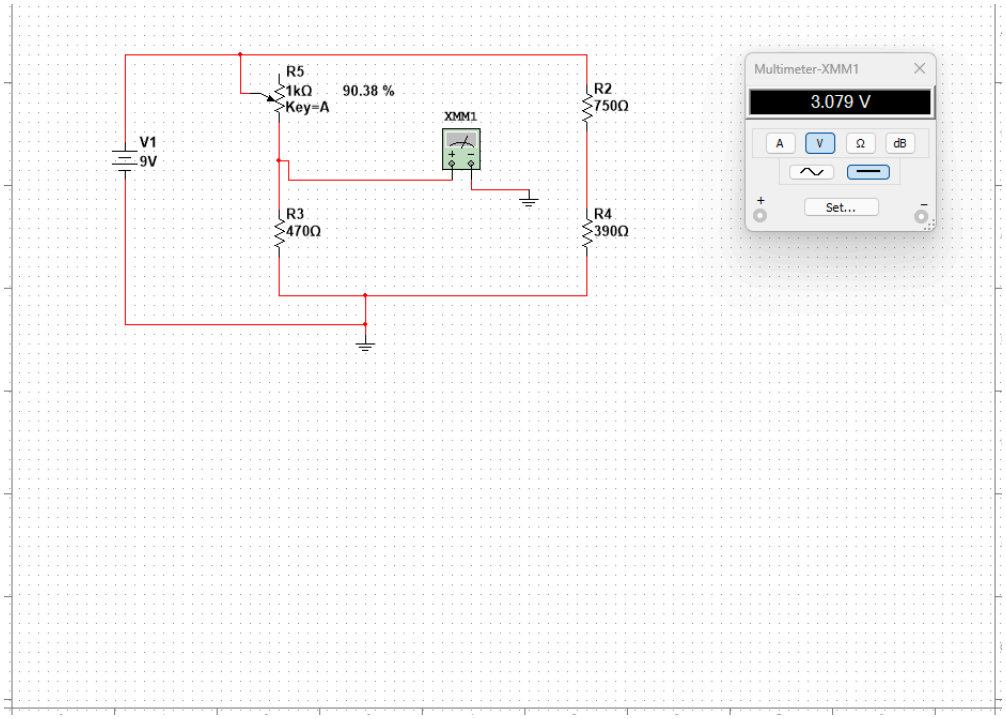


Figure 2. Balanced Wheatstone Bridge measuring V_A

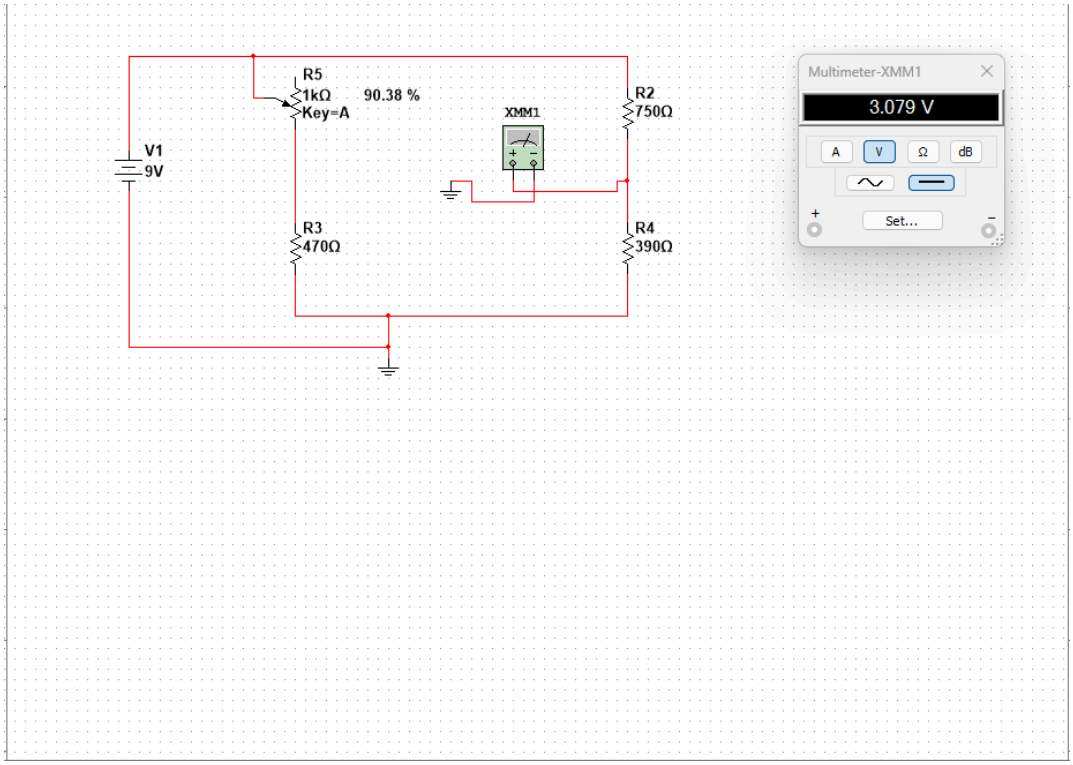


Figure 3. Balanced Wheatstone Bridge measuring V_B

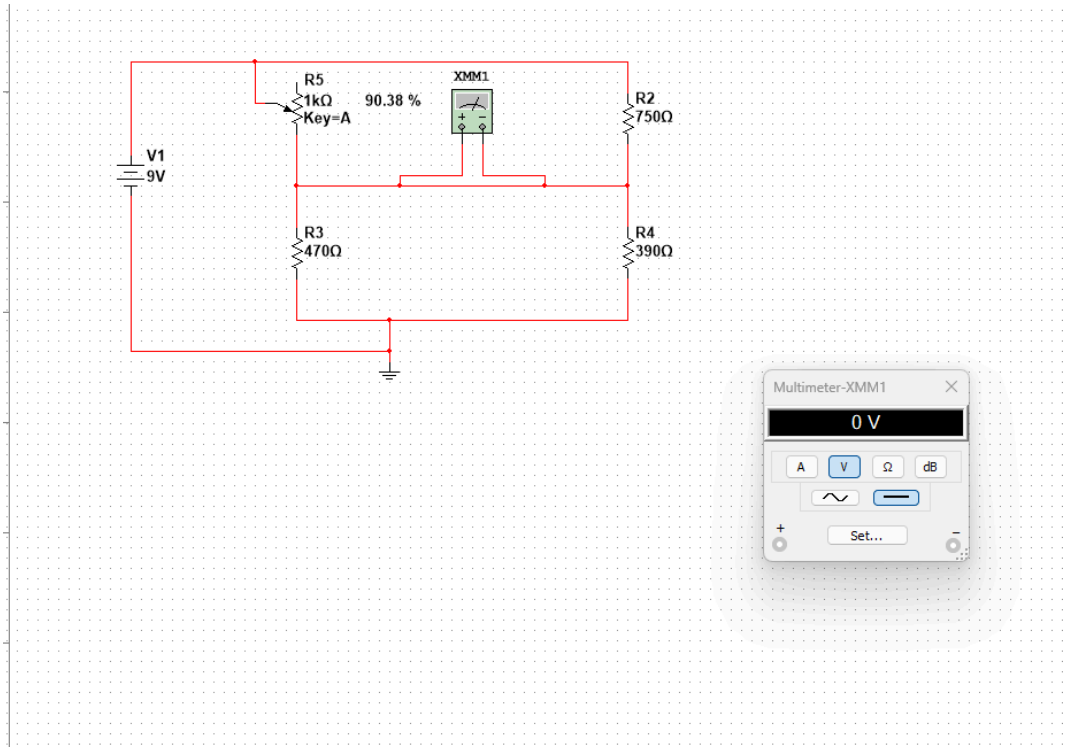


Figure 4. Balanced Wheatstone Bridge measuring V_OUT (no load)

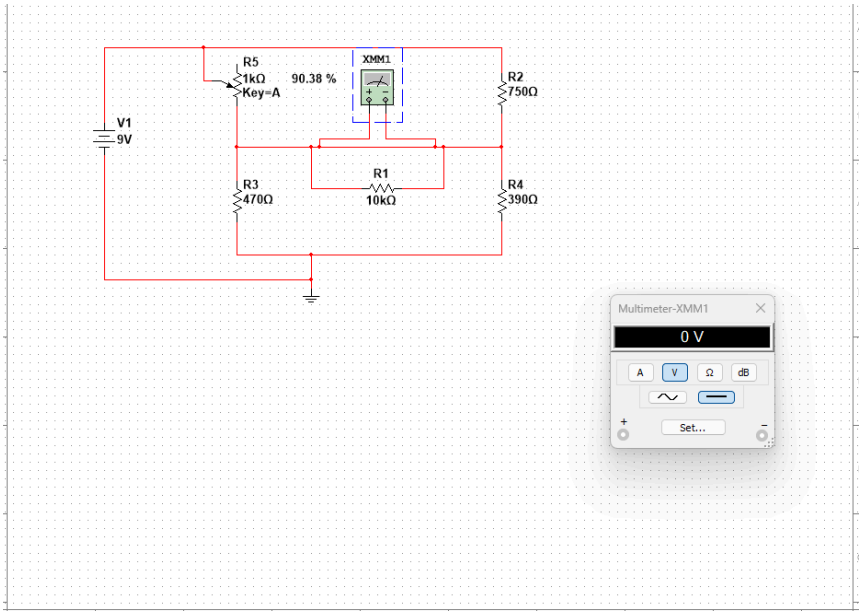


Figure 5. Balanced Wheatstone Bridge measuring V_OUT with 10 kΩ load connected

Part III. Circuit construction on the breadboard

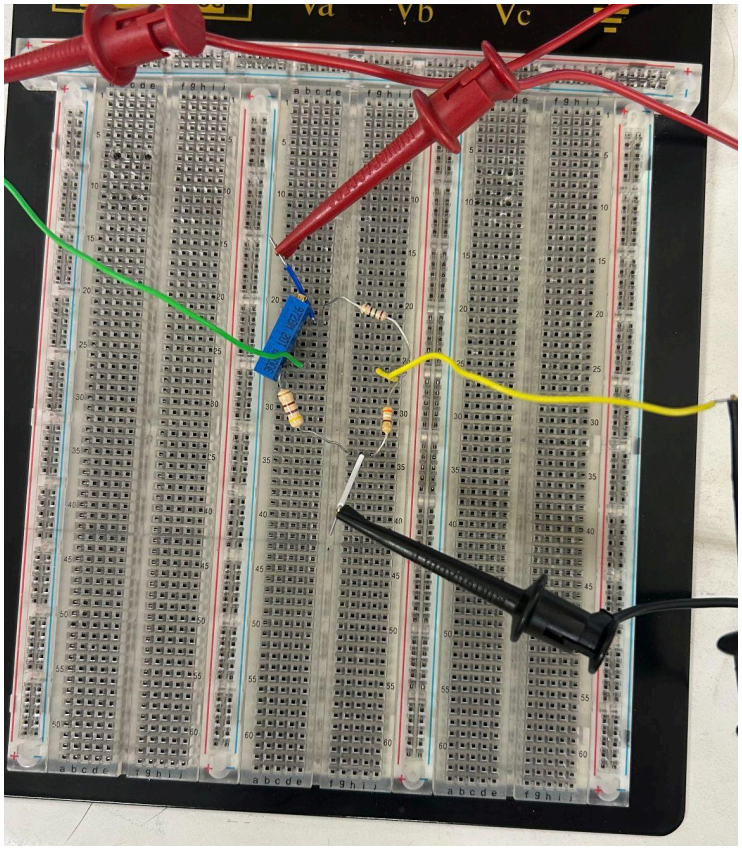


Figure 6. Balanced Wheatstone bridge measuring V_out

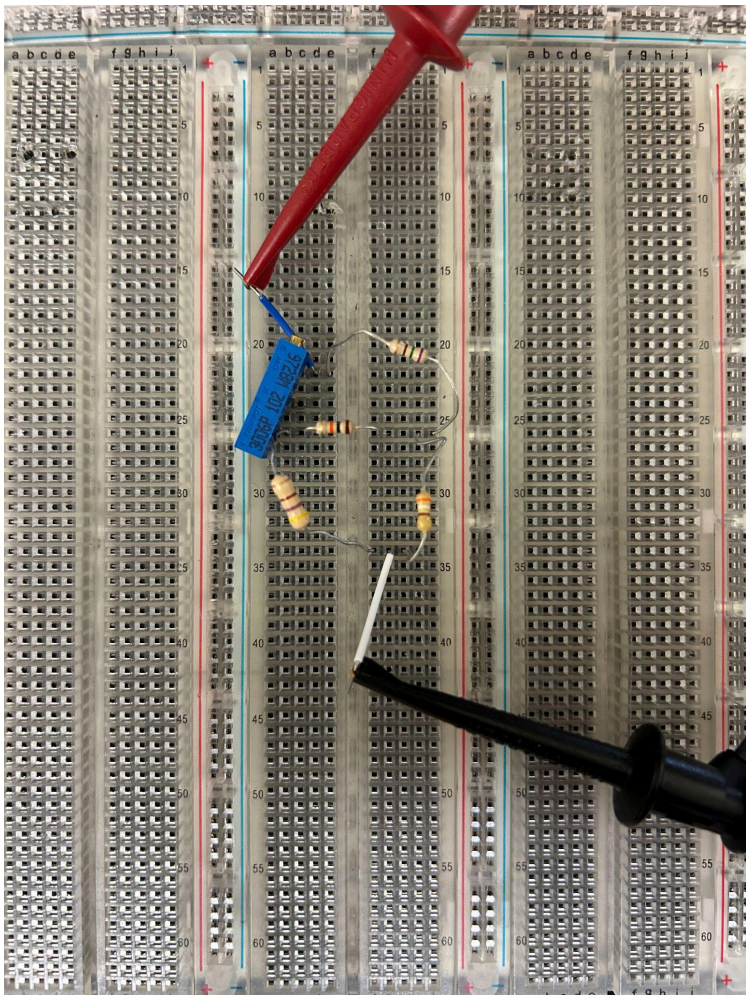


Figure 7. Balanced Wheatstone bridge with 10k load connected between points A and B.

Part IV. Questions

1. Compare the measured resistor values to the expected values. If they are not exactly the same then explain briefly why they are not exactly equal.

The measured values were close to the expected values, but not exactly the same. R_2, R_3, and R_4 measured at 746Ω, 469Ω, and 284Ω, respectively, which were slightly lower than the expected values of 750Ω, 470Ω, and 290Ω. The differences are mainly due to the manufactured tolerance of the resistors. Also, the potentiometer was the hardest to set and find an exact value due to the size of the component. Other causes could have been DMM accuracy and jumper wire resistance.

2. What do you observe regarding the relationship between the resistor ratios in the balanced bridge?

Compared to last week's lab with an unbalanced bridge, the ratios of the resistors are equal. R_1/R_3 and R_2/R_4 both result in a ratio of 1.923. This allows for V_A and V_B to be almost identical even when measuring breadboard values. V_OUT becomes 0 since the

wheatstone bridge is balanced.

3. What do you observe regarding the voltage between points A and B in the circuit when the bridge is balanced?

The voltage between points A and B when the circuit is balanced is the same, so the voltage between them is 0V. This is true for all the hand calculations, Multisim, and breadboard (with small percent error). When we connected the 10kΩ resistor and measured the V_out in that way, the value was still close to 0V at 0.016 V.

4. Compare the measured and the Multisim values of voltages and explain any differences.

Multisim gave values of V_A = 30.79 V and V_B = 30.79 V, and V_Out = 0 V, which matched the hand calculations exactly. The breadboard values were V_A = 3.145V and V_B = 3.011 V, which are 2.14% and 2.21% off from the calculated values. This is once again due to circuit resistance and resistor tolerance values.

5. From your experimental results have you gained a good understanding of balanced Wheatstone Bridge circuit?

Yes, we gained a good understanding of the balanced Wheatstone Bridge. We calculated the R_POT to be 903.85Ω to balance the bridge, found the same exact value in Multisim, and then tested it on a true breadboard circuit and got within a very close percent error. We followed the schematic of the Wheatstone circuit so that troubleshooting was easily possible.

Table 1. Resistance Data

	R_POT	**R_2**	**R_3**	**R_4**
Expected value	903.85 Ω	750 Ω	470 Ω	390 Ω
Measured	929 Ω	746 Ω	469 Ω	384 Ω
%Error Between Expected Value and Measured	2.78%	0.53%	0.21%	1.54%

Table 2. Balanced Bridge Data

	V_A	**V_B**	**V_OUT**	**V_OUT, Loaded**
Hand Calculations	3.079 V	3.079 V	0 V	N/A
Multisim	3.079 V	3.079 V	0 V	0 V
Breadboard	3.145 V	3.011 V	0.188 V	0.016 V
%Error Between Hand Calculations and Breadboard	2.14%	2.21%	N/A	N/A

Percent Error Calculations
%Error = |Measured - Expected| / Expected × 100%

Challenges:

One of the challenges we faced during this lab was adjusting the potentiometer to balance the Wheatstone Bridge circuit. It was difficult to get the exact resistance value because the measured resistors were slightly different from their expected values. We had to change multiple resistors to get the right measurement, and the 1K resistor didn't have a third leg. So we had to get the measurements correct and eventually solved the problem. Another challenge was connecting the components correctly on the breadboard and measuring the voltages accurately. We also noticed that the breadboard results were slightly different from the Multisim simulation, which may have been caused by resistor tolerances, loose connections, or measurement errors.

Conclusion:

In this lab, we learned how a balanced Wheatstone Bridge circuit works and how resistor ratios affect the voltage between two points. Through hand calculations and Multisim, we found that the voltage at points A and B was 3.079 V, resulting in an output voltage of 0 V. However, our breadboard measurements showed 3.145 V at point A and 3.011 V at point B, with a measured output voltage of 0.188 V. We also learned how to set up the breadboard after facing multiple mistakes and ended up finding the solution. This involved using the jumper cables to get the measurements right. These differences showed how real circuit components can affect the results. Overall, this lab helped us better understand how to balance a Wheatstone Bridge and compare theoretical calculations, simulations, and actual measurements.